SBE37-SMP-ODO MicroCAT

Instrument Configuration

Instrument Serial Number:	37-16246
Instrument Firmware Version:	2.13.0
Zero Conductivity Frequency:	2817.33
Communications Format:	RS232
Communications Settings:	9600 baud, 8 Data Bits, No Parity

Installed Devices/Sensors

Data Format	*Measurement*	*Sensor Type*	*Serial Number*	*Rating*
Count	Temperature	Internal	N/A	N/A
Frequency	Conductivity	Internal	N/A	N/A
Count	Pressure Sensor	Druck	10746413	2000m(2000 dBar)
RS232	Oxygen	SBE 63	63-1756	7000m

Maximum Depth: **2000m**

CAUTION - The maximum deployment depth will be limited by the measurement range of the pressure sensor, if installed, an attached sensor, if installed, or the housing.

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SENSOR SERIAL NUMBER: 16246
CALIBRATION DATE: 06-Oct-17

SBE 37 V2 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

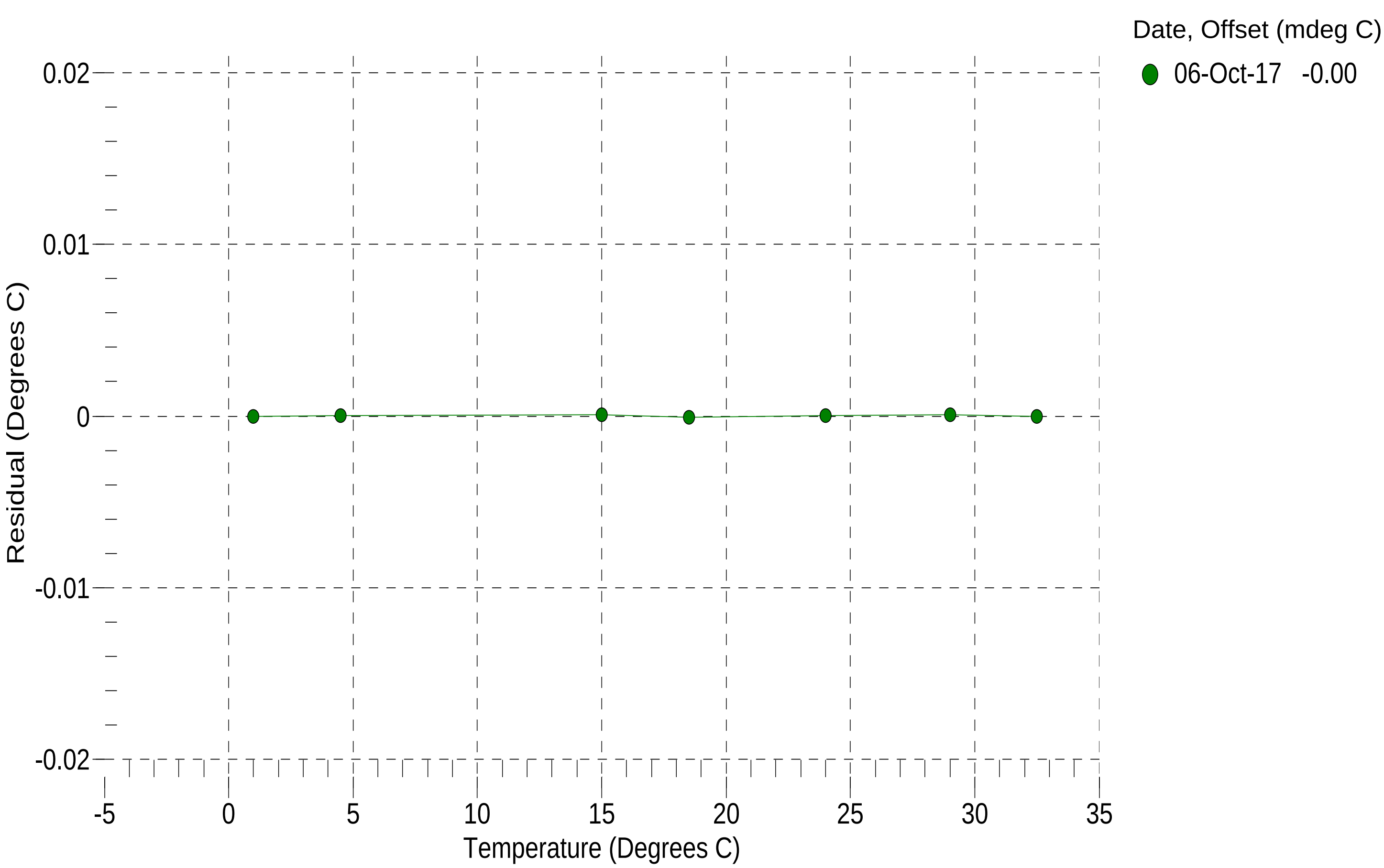
COEFFICIENTS:
a0 = -1.697941e-004
a1 = 3.191086e-004
a2 = -5.250845e-006
a3 = 2.198998e-007

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	567231.3	1.0000	-0.0000
4.5000	485422.5	4.5000	0.0000
15.0000	310268.7	15.0001	0.0001
18.5000	268944.5	18.4999	-0.0001
24.0000	216128.9	24.0000	0.0000
29.0000	178261.0	29.0001	0.0001
32.5000	156297.6	32.5000	-0.0000

n = Instrument Output (counts)

Temperature ITS-90 (°C) = $1/\{a0 + a1[\ln(n)] + a2[\ln^2(n)] + a3[\ln^3(n)]\} - 273.15$

Residual (°C) = instrument temperature - bath temperature



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SENSOR SERIAL NUMBER: 16246
CALIBRATION DATE: 06-Oct-17

SBE 37 V2 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

```
g = -9.754290e-001
h =  1.229602e-001
i = -8.344709e-005
j =  2.091873e-005

CPcor = -9.5700e-008
CTcor =  3.2500e-006
WBOTC = -1.1830e-007
```

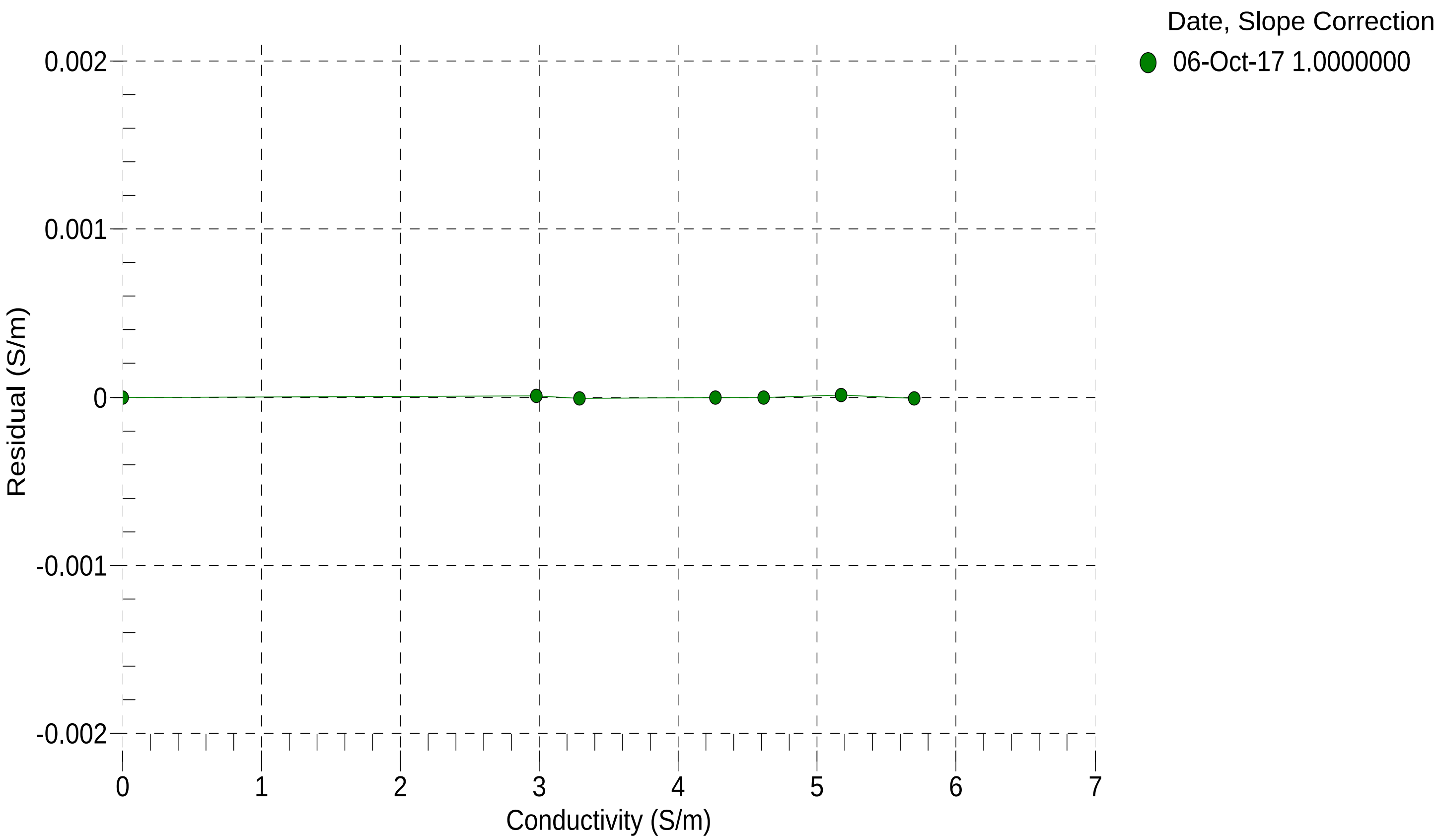
BATH TEMP (° C)	BATH SAL (PSU)	BATH COND (S/m)	INSTRUMENT OUTPUT (Hz)	INSTRUMENT COND (S/m)	RESIDUAL (S/m)
22.0000	0.0000	0.00000	2817.33	0.00000	0.00000
1.0000	34.8645	2.97969	5666.93	2.97970	0.00001
4.5000	34.8444	3.28711	5882.25	3.28711	-0.00001
15.0000	34.8011	4.26995	6522.37	4.26994	-0.00000
18.5000	34.7920	4.61549	6732.70	4.61549	-0.00000
24.0000	34.7819	5.17407	7059.17	5.17409	0.00001
29.0000	34.7764	5.69652	7351.08	5.69651	-0.00001
32.5000	34.7735	6.06937	7551.68	6.06816	-0.00121

f = Instrument Output(Hz) * sqrt(1.0 + WBOTC * t) / 1000.0

t = temperature (°C); p = pressure (decibars); δ = CTcor; ε = CPcor;

Conductivity (S/m) = $(g + h * f^2 + i * f^3 + j * f^4) / (1 + \delta * t + \varepsilon * p)$

Residual (Siemens/meter) = instrument conductivity - bath conductivity



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SENSOR SERIAL NUMBER: 16246
CALIBRATION DATE: 29-Sep-17

SBE 37 V2 PRESSURE CALIBRATION DATA
2900 psia S/N 10746413

COEFFICIENTS:

```
PA0 =       4.446122e-001
PA1 =       8.977838e-003
PA2 =      -1.546084e-010
PTEMPA0 = -5.980709e+001
PTEMPA1 =  5.426103e-002
PTEMPA2 = -6.168675e-007

PTCA0 =  5.238001e+005
PTCA1 =  3.284043e+000
PTCA2 = -4.550586e-002
PTCB0 =  2.519075e+001
PTCB1 =  3.491272e-004
PTCB2 =  0.000000e+000
```

PRESSURE SPAN CALIBRATION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (volts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)
14.59	525429.0	1545.3	14.61	0.00
591.74	589811.1	1551.1	591.76	0.00
1168.89	654328.0	1551.6	1168.84	-0.00
1746.05	719004.2	1552.3	1746.05	0.00
2323.18	783816.7	1552.9	2323.19	0.00
2900.25	848762.5	1553.9	2900.20	-0.00
2323.17	783828.4	1552.9	2323.29	0.00
1746.16	719007.8	1552.5	1746.08	-0.00
1168.93	654341.3	1551.6	1168.96	0.00
591.68	589801.3	1551.9	591.67	-0.00
14.59	525425.6	1551.8	14.57	-0.00

THERMAL CORRECTION

TEMP (°C)	THERMISTOR OUTPUT (volts)	INSTRUMENT OUTPUT (counts)
32.50	1735	525516.00
29.00	1668	525518.55
24.00	1573	525514.30
18.50	1467	525503.97
15.00	1401	525495.99
4.50	1202	525474.84
1.00	1135	525462.26

TEMPERATURE (°C)	SPAN (mV)
-5.00	25.19
35.10	25.20

y = thermistor output (counts)

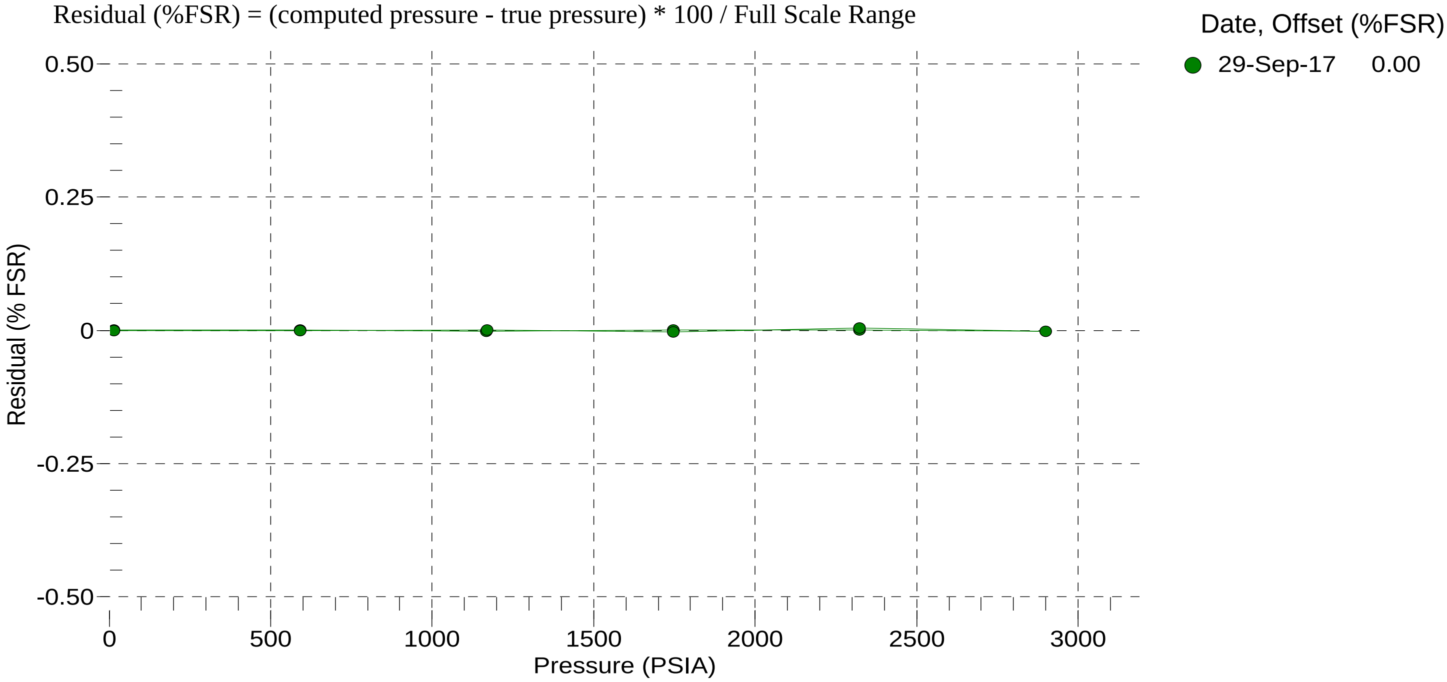
$t = PTEMPA0 + PTEMPA1 * y + PTEMPA2 * y^2$

$x = \text{instrument output} - PTCA0 - PTCA1 * t - PTCA2 * t^2$

$n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)$

$\text{pressure (PSIA)} = PA0 + PA1 * n + PA2 * n^2$

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range



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Pressure Test Certificate

Test Date: 2017-09-28

Description: SBE-37 Microcat

Sensor Information:

Model Number: SBE-37

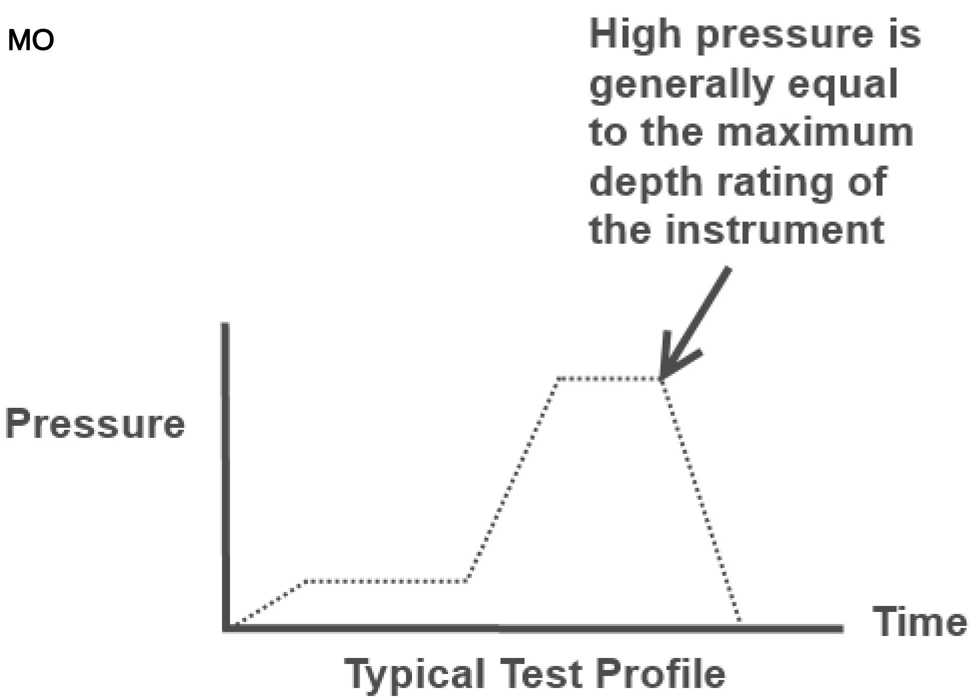
Serial Number: 16246

Pressure Test Protocol:

Low Pressure Test:	40	PSI	Held For:	15	Minutes
High Pressure Test:	2900	PSI	Held For:	15	Minutes

Passed Test: True

Tested By: MO



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SENSOR SERIAL NUMBER: 1756
CALIBRATION DATE: 20-Sep-17

SBE 63 OXYGEN TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

TA0 = 7.537160e-004 TA2 = 2.199087e-006
TA1 = 2.355728e-004 TA3 = 5.267960e-008

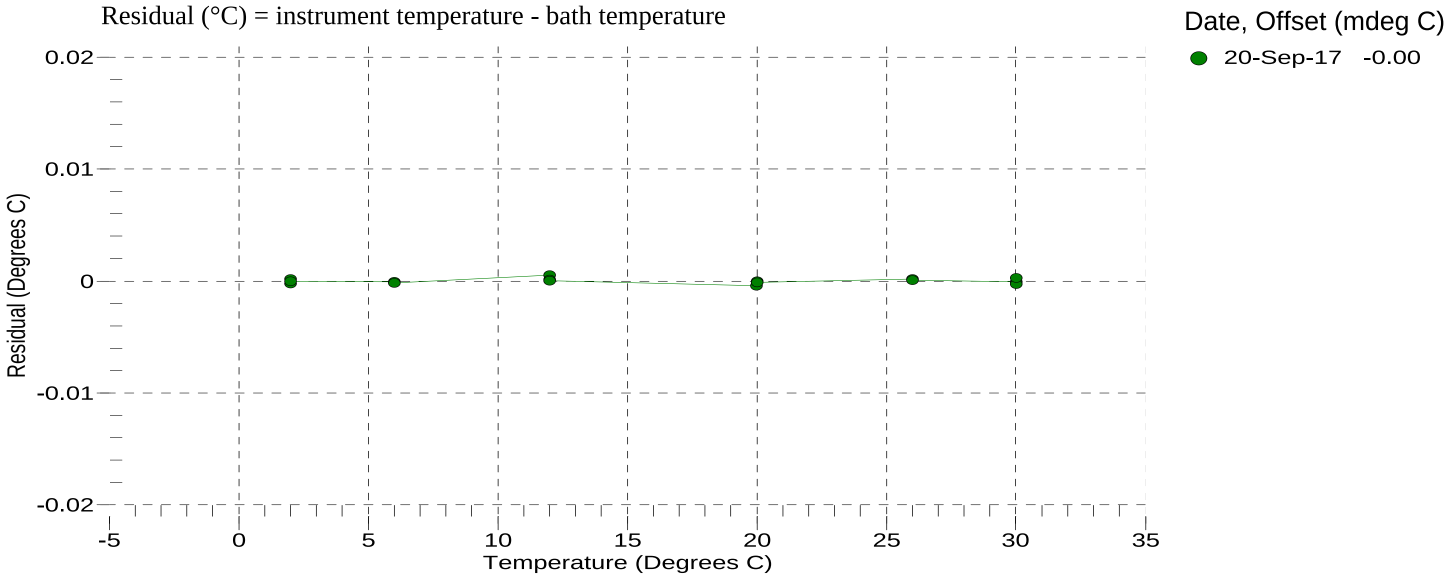
BATH TEMP (° C)	INSTRUMENT OUTPUT(V)	INST TEMP (° C)	RESIDUAL (° C)
1.9999	1.11867	2.0001	0.00019
1.9999	1.11867	2.0001	0.00019
2.0000	1.11868	1.9998	-0.00022
2.0001	1.11867	2.0001	-0.00001
6.0000	0.99458	5.9999	-0.00007
6.0000	0.99458	5.9999	-0.00007
6.0001	0.99458	5.9999	-0.00017
6.0001	0.99458	5.9999	-0.00017
12.0000	0.82887	12.0005	0.00051
12.0000	0.82888	12.0001	0.00012
12.0001	0.82888	12.0001	0.00002
12.0001	0.82888	12.0001	0.00002
19.9999	0.64539	19.9995	-0.00042
20.0000	0.64538	20.0000	-0.00003
20.0001	0.64538	20.0000	-0.00013
20.0001	0.64538	20.0000	-0.00013
25.9999	0.53346	26.0001	0.00019
25.9999	0.53346	26.0001	0.00019
26.0000	0.53346	26.0001	0.00009
26.0000	0.53346	26.0001	0.00009
30.0000	0.46968	29.9999	-0.00009
30.0000	0.46968	29.9999	-0.00009
30.0002	0.46968	29.9999	-0.00029
30.0003	0.46967	30.0006	0.00028

V = Instrument Output (Volts)

L = *ln* (100000 * V / (3.3 - V))

Temperature ITS-90 (°C) = 1 / (TA0 + (TA1 * L) + (TA2 * L^2) + (TA3 * L^3)) - 273.15

Residual (°C) = instrument temperature - bath temperature



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CALIBRATION DATE: 20-Sep-17

SBE 63 OXYGEN CALIBRATION DATA

COEFFICIENTS:

A0 = 1.0513e+000	B0 = -2.4911e-001	C0 = 1.0363e-001	E = 1.1000e-002
A1 = -1.5000e-003	B1 = 1.6349e+000	C1 = 4.4103e-003	
A2 = 4.0281e-001		C2 = 6.0736e-005	

BATH OXYGEN (ml/l)	BATH TEMPERATURE (° C)	BATH SALINITY (PSU)	INSTRUMENT OUTPUT (µsec)	INSTRUMENT OXYGEN (ml/l)	RESIDUAL (ml/l)
0.714	30.00	0.00	31.16	0.712	-0.002
0.732	26.00	0.00	31.89	0.731	-0.001
0.788	20.00	0.00	32.80	0.790	0.002
0.874	12.00	0.00	34.17	0.874	0.001
0.980	6.00	0.00	35.07	0.983	0.003
1.077	2.00	0.00	35.62	1.084	0.006
2.190	30.00	0.00	22.85	2.188	-0.003
2.325	26.00	0.00	23.39	2.324	-0.001
2.468	20.00	0.00	24.53	2.470	0.002
2.982	12.00	0.00	25.43	2.978	-0.004
3.396	6.00	0.00	26.38	3.391	-0.005
3.643	30.00	0.00	18.90	3.640	-0.002
3.749	2.00	0.00	27.01	3.742	-0.007
3.899	26.00	0.00	19.32	3.901	0.001
4.319	20.00	0.00	20.05	4.327	0.007
5.071	12.00	0.00	21.10	5.067	-0.004
5.194	30.00	0.00	16.36	5.188	-0.006
5.607	26.00	0.00	16.67	5.609	0.002
5.815	6.00	0.00	21.93	5.810	-0.005
6.221	20.00	0.00	17.32	6.229	0.008
6.432	2.00	0.00	22.52	6.432	0.001
7.286	12.00	0.00	18.27	7.284	-0.002
8.344	6.00	0.00	19.05	8.345	0.001
8.858	2.00	0.00	19.91	8.863	0.005

T = temperature (°C) , P = pressure (dbar), U = Instrument output (µsec)

S_{corr} (salinity correction function) = 1.0 for calibration in DI water

See the user manual for more information on S_{corr} calculation

V = U / 39.457071

Oxygen (ml/l) = $\{((A0 + A1*T + A2 * V^2)/(B0 + B1 * V) - 1.0)/(C0 + C1 * T + C2 * T^2)\} * S_{corr} * \exp(E * P / (T + 273.15))$

Residual (ml/l) = instrument oxygen - bath oxygen